| | | | | |
|---|---|---|---|---|
| **4** | **(a)** | displacement (directly) proportional to acceleration/force | M1 | |
| | | *either* displacement and acceleration in opposite directions<br>*or* acceleration (always) towards a (fixed) point | A1 | [2] |

**(b) (i)** ⅓π rad or 1.05 rad (*allow 60° if unit clear*) A1 [1]

**(ii)** $a_0 = -\omega^2 x_0$
$= (-)\ (2\pi / 1.2)^2 \times 0.030$ C1
$= (-)\ 0.82\,\mathrm{m\,s^{-2}}$ A1 [2]
(*special case: using oscillator P gives $x_0$ = 1.7 cm and $a_0$ = 0.47 m s$^{-1}$ for 1/2*)

**(iii)** max. energy $\propto x_0^2$
ratio $= 3.0^2 / 1.7^2$ C1
$= 3.1$ (*at least 2 s.f.*) A1 [2]
(*if has inverse ratio but has stated max. energy $\propto x_0^2$ then allow 1/2*)

**(c)** graph: straight line through (0,0) with negative gradient M1
correct end-points (–3.0, +0.82) and (+3.0, –0.82) A1 [2]